

Deoxyhypusine Synthase (DHPS) Deficiency — Comprehensive Disease Characterization Report

MONDO ID: MONDO:0032775 | **OMIM phenotype:** #618480 | **Category:** Genetic (autosomal recessive neurodevelopmental disorder)

Summary

Deoxyhypusine synthase (DHPS) deficiency is an **ultra-rare autosomal recessive neurodevelopmental disorder** caused by biallelic hypomorphic (partial loss-of-function) variants in the *DHPS* gene on chromosome 19p13.13. It is one of five recognized "**polyaminopathies**" — inborn disorders of polyamine biosynthesis/utilization — and belongs to the subgroup of **hypusination (eIF5A) disorders**. DHPS catalyzes the first and committed step in the post-translational synthesis of the unique amino acid **hypusine**, which is present in only one cellular protein, **eukaryotic translation initiation factor 5A (eIF5A)** and its homolog eIF5A2. Hypusinated eIF5A is required for the efficient translation-elongation of mRNAs containing polyproline stretches and for general control of ribosomal elongation and termination. Because complete loss of DHPS is embryonic-lethal in mice, all surviving human patients carry **partial-function allele combinations**.

The disorder was first delineated by Ganapathi et al. in 2019 (*American Journal of Human Genetics*), who described five affected individuals from four unrelated families. Remarkably, all five shared a **recurrent hypomorphic missense allele, c.518A>G (p.Asn173Ser)**, in *trans* with a near-null allele (a splice-site, in-frame deletion, or start-loss variant). The clinical picture is a **static (non-progressive) encephalopathy** dominated by global developmental delay/intellectual disability and seizures, with variable hypotonia, microcephaly, short stature, and mild facial dysmorphism. Model organisms — conditional/neuron-specific *Dhps* knockout mice and *dhps* mutant zebrafish — recapitulate the growth, cognitive, and epilepsy phenotypes, and the zebrafish model provides a circuit-level mechanism via **reduced arborization of inhibitory (GABAergic) interneurons**.

Diagnosis is molecular — via whole-exome or whole-genome sequencing — and no validated metabolite biomarker or newborn-screening assay exists. Management is currently **supportive/symptomatic** (anti-seizure medication, developmental therapies). A preclinical **AAV-based gene therapy** for DHPS syndrome was reported in 2026, representing the first disease-modifying strategy in development. Because only ~5 patients have been fully characterized, formal epidemiology, penetrance, expressivity, and natural-history data remain undefined, and much mechanistic understanding is extrapolated from model systems and the closely related aging/spermidine literature.

Key Findings

Finding 1 — DHPS deficiency is an autosomal recessive neurodevelopmental disorder caused by biallelic hypomorphic *DHPS* variants

Ganapathi et al. (2019) identified rare biallelic, recurrent, predicted likely-pathogenic *DHPS* variants segregating with disease in **five affected individuals from four unrelated families**. All five affected individuals share a recurrent missense variant, **c.518A>G (p.Asn173Ser)**, in *trans* with a likely gene-disrupting variant (c.1014+1G>A splice-site; c.912_917delTTACAT [p.Tyr305_Ile306del] in-frame deletion; or c.1A>G [p.Met1?] start-loss). Functional testing of recombinant enzyme demonstrated that the **p.Asn173Ser** protein retained only ~20% of normal in vitro activity, while **p.Tyr305_Ile306del** had absent activity — consistent with a hypomorphic loss-of-function mechanism.

The genotype architecture is important: because **constitutive *Dhps*-null mice are embryonic lethal**, complete loss of DHPS function is presumed incompatible with human life. Surviving patients therefore invariably carry a **partial-function (hypomorphic) allele** — the relatively common p.Asn173Ser — that provides enough residual activity to permit development, paired in *trans* with a more severe (near-null) allele.

"we identified rare biallelic, recurrent, predicted likely pathogenic variants in DHPS segregating with disease in five affected individuals from four unrelated families. These individuals have similar neurodevelopmental features that include global developmental delay and seizures" — PMID: 30661771

"Recombinant DHPS enzyme harboring either the p.Asn173Ser or p.Tyr305_Ile306del variant showed reduced (20%) or absent in vitro activity" — PMID: 30661771

Evidence type: Human clinical + in vitro enzymology. **Inheritance:** Autosomal recessive.

Finding 2 — Mechanism: DHPS catalyzes the first step of hypusine synthesis on eIF5A, essential for translation of polyproline motifs

Hypusine [N-ε-(4-amino-2-hydroxybutyl)lysine] is a unique amino acid formed post-translationally from lysine and is found in a single cellular protein — **eIF5A** — and its homolog **eIF5A2**. Its biosynthesis is a **two-step reaction**: (1) **DHPS** transfers the aminobutyl moiety of the polyamine **spermidine** to a specific conserved lysine residue of eIF5A, forming the intermediate **deoxyhypusine**; then (2) **deoxyhypusine hydroxylase (DOHH)** hydroxylates deoxyhypusine to form mature hypusine. DHPS thus performs the committed, rate-defining first step.

Hypusinated eIF5A functions principally in **translation ELONGATION**: it facilitates the translation of peptide sequences containing **polyproline stretches** and exerts a broad regulatory effect on the elongation and termination phases of protein synthesis. DHPS is highly conserved and essential for eukaryotic life.

"Hypusine is formed post-translationally from lysine and is found in a single cellular protein, eukaryotic translation initiation factor-5A (eIF5A), and its homolog eIF5A2. Biosynthesis of hypusine is a two-step reaction involving the enzymes deoxyhypusine synthase (DHPS) and deoxyhypusine hydroxylase (DOHH)" — PMID: 30661771

"eIF5A facilitates the translation of peptide sequences containing polyproline stretches and exerts a universal regulatory effect on the elongation and termination phases of protein synthesis" — PMID: 39303786

Ontology suggestions: GO:0008612 (peptidyl-lysine modification to peptidyl-hypusine); GO:0006414 (translational elongation); CHEBI:16610 (spermidine); CHEBI:59905 (hypusine).

Finding 3 — Downstream mechanism: hypusinated eIF5A supports neuronal mitochondrial function and autophagy; spermidine is the substrate

Beyond bulk protein synthesis, independent studies show that eIF5A hypusination is required for **mitochondrial respiratory competence and autophagy in neurons**. Dietary **spermidine** — the aminobutyl donor for the DHPS reaction — crosses the blood-brain barrier in mice, increases **hippocampal eIF5A hypusination and mitochondrial function**, and improves cognition in aged animals (Schroeder et al., 2021). A parallel study demonstrated that spermidine-induced hypusination **preserves mitochondrial and cognitive function during aging** (Hofer et al., 2021), with effects dependent on autophagy/mitophagy machinery (Atg7, Pink1/Parkin).

While these studies were conducted in aging (not DHPS-deficiency) models, they establish the mechanistically relevant **hypusination → mitochondrial function → cognition** axis. In DHPS deficiency, reduced hypusination is inferred to compromise this axis in developing neurons, contributing to the neurodevelopmental phenotype. This downstream link is **inferred by analogy** rather than directly demonstrated in patient tissue.

"dietary spermidine passes the blood-brain barrier in mice and increases hippocampal eIF5A hypusination and mitochondrial function" — PMID: 33852843

"Spermidine-induced hypusination preserves mitochondrial and cognitive function during aging" — PMID: 34105442

Ontology suggestions: GO:0006914 (autophagy); GO:0045333 (cellular respiration); GO:0005739 (mitochondrion); CHEBI:16610 (spermidine).

Finding 4 — Clinical phenotype: global developmental delay, seizures, hypotonia, microcephaly, short stature, dysmorphism

The **cardinal features** in the founding cohort (5 patients / 4 families) were **global developmental delay / intellectual disability (5/5, 100%)** and **seizures/epilepsy** (present in the majority). **Short stature** was noted in 2 of 4 affected females. Broader reviews of the hypusination disorders (DHPS/DOHH/EIF5A) describe a shared spectrum comprising **prenatal issues, hypotonia, dysmorphisms, microcephaly, moderate-to-severe**

neurodevelopmental disorder/intellectual disability, and behavioral disorders. The zebrafish-model summary characterizes the human syndrome as causing "epilepsy, cognitive and motor impairments, and mild facial dysmorphology."

Onset is **congenital/infantile**, and the course is **chronic and non-progressive** (a static encephalopathy) with lifelong disability.

Phenotype	HPO term	Frequency (founding cohort)	Type
Global developmental delay	HP:0001263	5/5 (100%)	Neurodevelopmental
Intellectual disability	HP:0001249	High (majority–all)	Neurodevelopmental
Seizures / epilepsy	HP:0001250	Majority	Clinical sign / neurological
Hypotonia	HP:0001252	Reported	Clinical sign
Microcephaly	HP:0000252	Reported	Physical manifestation
Short stature	HP:0004322	2/4 females	Physical manifestation
Facial dysmorphism	HP:0001999	Reported (mild)	Physical manifestation
Behavioral abnormality	HP:0000708	Reported	Behavioral

"These individuals have similar neurodevelopmental features that include global developmental delay and seizures" — PMID: 30661771

"Two of four affected females have short stature" — PMID: 30661771

"Main phenotypic features consisted of prenatal issues, hypotonia, dysmorphisms, microcephaly, moderate-severe neurodevelopmental disorders/intellectual disability and behavioral disorders" — PMID: 40883692

Finding 5 — DHPS deficiency is one of five polyaminopathies; a hypusination/eIF5A disorder with emerging gene therapy

The **polyaminopathies** comprise five rare neurodevelopmental disorders that disrupt polyamine biosynthesis/utilization:

Disorder	Gene	Mechanism	Inheritance	First described
Snyder-Robinson syndrome	<i>SMS</i>	Loss of function	X-linked	1969
Bachmann-Bupp syndrome	<i>ODC1</i>	Gain of function	AD (de novo)	~past 7 yr
Faundes-Banka syndrome	<i>EIF5A</i>	Loss of function	AD	~past 7 yr
DHPS deficiency	<i>DHPS</i>	Biallelic hypomorphic LoF	AR	2019
DOHH disorder	<i>DOHH</i>	Biallelic LoF	AR	~past 7 yr

Four of the five (including DHPS deficiency) have been identified only within roughly the past seven years, underscoring their recency and rarity. Treatment is currently **supportive/symptomatic** (anti-seizure medications and developmental therapies). Notably, a **preclinical AAV gene-therapy strategy** for DHPS syndrome was reported in 2026 (Santo et al.), representing the first disease-modifying approach in development.

"DHPS (deoxyhypusine synthase) deficiency is an autosomal recessive disease and results from bi-allelic hypomorphic variants in the deoxyhypusine synthase (DHPS) gene, which results in reduced deoxyhypusine synthase enzyme activity" — PMID: 41410504

"Snyder-Robinson syndrome was first described in 1969, while the other four syndromes have only been identified in the past 7 years" — PMID: 41410504

"Deoxyhypusine synthase (DHPS) syndrome is a rare, autosomal recessive neurodevelopmental disorder caused by biallelic pathogenic variants" — PMID: 42239796

Ontology suggestion: NCIT — gene therapy / AAV vector-based gene transfer.

Finding 6 — Gene identifiers, locus, and allele frequency of the recurrent p.Asn173Ser hypomorphic variant

Gene identifiers: NCBI Gene ID **1725**; HGNC:**2869**; OMIM gene **600944**; UniProt **P49366**; Ensembl **ENSG00000095059**. **Locus:** chromosome **19p13.13** (GRCh38 chr19:12,673,411–12,681,901, minus strand). **Disease phenotype:** OMIM **#618480**.

Querying gnomAD v4, the recurrent hypomorphic missense **c.518A>G (p.Asn173Ser)** has an exome allele frequency of $\approx 9.9 \times 10^{-5}$ and a genome AF of $\approx 5.3 \times 10^{-5}$ — rare but recurrent, consistent with a tolerated partial-function allele carried heterozygously in the general population. Other codon-173 variants exist at low frequency (e.g., p.Asn173Lys, AF $\approx 4.9 \times 10^{-5}$).

Variant classes documented in patients: **missense** (p.Asn173Ser), **splice-site** (c.1014+1G>A), **in-frame deletion** (p.Tyr305_Ile306del), and **start-loss** (c.1A>G, p.Met1?). All are **germline**; the functional consequence is **partial loss of function (hypomorphic)** — complete loss of function is embryonic-lethal.

"All five affected individuals share a recurrent missense variant (c.518A>G [p.Asn173Ser]) in trans with a likely gene disrupting variant (c.1014+1G>A, c.912_917delTTACAT [p.Tyr305_Ile306del], or c.1A>G [p.Met1?])" — PMID: 30661771

Ontology suggestions: SO:0001583 (missense variant); SO:0001574 (splice-acceptor/donor variant); SO:0001822 (inframe deletion); SO:0002012 (start-lost).

Finding 7 — Epidemiology, diagnosis, and prognosis: ultra-rare, molecularly diagnosed, static lifelong course

Epidemiology: Ultra-rare. The founding report described only **5 affected individuals from 4 families**, and subsequent reviews note only ~5 individuals characterized to date. Formal prevalence and incidence are **not established**; Orphanet lists it among ultra-rare disorders. Inheritance is **autosomal recessive** with presumed complete penetrance in biallelic carriers; **both sexes** are affected. Consanguinity and founder effects are not established — recurrence is driven by the relatively common **p.Asn173Ser** hypomorphic allele rather than a founder haplotype.

Diagnosis: Molecular, via **whole-exome or whole-genome sequencing** (or *DHPS* single-gene / NDD-epilepsy panel testing). Confirmatory functional support comes from **reduced recombinant DHPS enzyme activity** and **impaired eIF5A hypusination** assays. There is **no validated blood/urine metabolite biomarker or newborn-screening assay**. Supportive workup: EEG (epileptiform activity), **brain MRI**, and developmental assessment.

Differential diagnosis: Other polyaminopathies (DOHH disorder, Faundes-Banka/*EIF5A*, Snyder-Robinson, Bachmann-Bupp) and other genetic developmental and epileptic encephalopathies.

Prognosis: Chronic, **static (non-degenerative) encephalopathy** with lifelong intellectual disability and epilepsy; no evidence of progressive neurodegeneration. Formal survival/mortality data are unavailable given the small cohort.

"we identified rare biallelic, recurrent, predicted likely pathogenic variants in *DHPS* segregating with disease in five affected individuals from four unrelated families" — [PMID: 30661771](#)

"one reporting 5 subjects with *DHPS*-related disorders (*DHPS-D*)" — [PMID: 40883692](#)

Finding 8 — Deep evolutionary conservation and validated model organisms

DHPS orthologs (HomoloGene 1453) span the eukaryotic tree, reflecting that hypusine/eIF5A modification is essential for eukaryotic life:

Species	Gene	NCBI Gene ID	Database
Human (<i>Homo sapiens</i>)	<i>DHPS</i>	1725	NCBI / HGNC
Mouse (<i>Mus musculus</i>)	<i>Dhps</i>	330817	MGI
Rat (<i>Rattus norvegicus</i>)	<i>Dhps</i>	288923	RGD
Zebrafish (<i>Danio rerio</i>)	<i>dhps</i>	406329	ZFIN
Fruit fly (<i>Drosophila melanogaster</i>)	<i>Dhps</i>	38917	FlyBase
Nematode (<i>C. elegans</i>)	<i>dhps-1</i>	174840	WormBase
Budding yeast (<i>S. cerevisiae</i>)	<i>DYS1</i>	856465	SGD

Validated disease models: - **Mouse** — constitutive *Dhps* knockout is **embryonic lethal**; **conditional neuron/brain-specific *Dhps* (or *Eif5a*) deletion** impairs growth, viability, neurodevelopment, and cognition (Kar et al., 2021), recapitulating the human developmental/cognitive deficits. - **Zebrafish** — the *dhps* mutant shows **aberrant morphology, epileptiform activity, and reduced arborization of inhibitory interneurons** (Shojaeinia et al., 2024), recapitulating the epilepsy phenotype and providing a circuit-level mechanism. - **Cellular/in vitro** — HEK293T co-transfection assays showed mutant DHPS reduces eIF5A hypusination (Ganapathi, 2019); iPSC/organoid and AAV gene-therapy models are emerging (Santo et al., 2026).

"Neuron-specific ablation of eIF5A or deoxyhypusine synthase leads to impairments in growth, viability, neurodevelopment, and cognitive functions in mice" — PMID: 34688659

"DHPS is also highly conserved and is essential for life, as Dhps-null mice are embryonic lethal" — PMID: 30661771

Finding 9 — Anatomical involvement: central nervous system (cortex and inhibitory interneuron circuits), with growth axis secondary

Although DHPS is **ubiquitously expressed** and hypusination occurs in all cells, the clinical phenotype is **dominated by the central nervous system**: global developmental delay, intellectual disability, and epilepsy. Zebrafish modeling localizes the dysfunction to **inhibitory (GABAergic) interneurons** — reduced arborization producing epileptiform activity — implicating cortical/forebrain inhibitory circuits. **Microcephaly** and **dysmorphic facial features** indicate effects on brain and craniofacial growth; **short stature** indicates a secondary effect on the somatic growth axis. Subcellularly, the primary defect is **cytoplasmic** (translation), with downstream impact on **mitochondria** and **autophagy/lysosomal turnover**.

"reduced arborization of inhibitory interneurons" — PMID: 39334388

Ontology suggestions: UBERON:0000955 (brain); UBERON:0000956 (cerebral cortex); CL:0000617 (GABAergic neuron / inhibitory interneuron); GO:0005737 (cytoplasm); GO:0005739 (mitochondrion).

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic hypomorphic DHPS variants** (recurrent p.Asn173Ser in *trans* with a near-null allele) **lead to** reduced DHPS enzyme protein activity (~20% residual for p.Asn173Ser; near-absent for null alleles). *[Demonstrated: in vitro enzymology,*

P 30661771

2. Reduced DHPS activity **results in** decreased transfer of the aminobutyl group from **spermidine** to eIF5A → reduced formation of **deoxyhypusine**, and consequently reduced mature **hypusine** on eIF5A. *[Demonstrated: cellular hypusination assay,*

P 30661771

3. Hypomodified (hypusine-deficient) eIF5A **leads to** impaired **translation elongation**, particularly of mRNAs encoding **polyproline motifs** and other elongation-sensitive transcripts. *[Demonstrated in general eIF5A biology;*

P 39303786

— *inferred for patient tissue]*

4. Impaired translation **results in** downstream deficits in **neuronal mitochondrial respiratory function and autophagy/mitophagy**. *[Inferred by analogy from spermidine/aging models, PMIDs 33852843, 34105442]*

5. **Branch A (neuronal circuits):** In developing brain, this **leads to** reduced arborization of **inhibitory (GABAergic) interneurons** → cortical excitation/inhibition imbalance → **epileptiform activity and seizures**. *[Demonstrated: zebrafish,*

P 39334388

6. **Branch B (neurodevelopment/cognition):** Impaired neuronal protein synthesis and viability **result in** **global developmental delay, intellectual disability, hypotonia, and microcephaly**. *[Demonstrated: conditional mouse,*

P 34688659

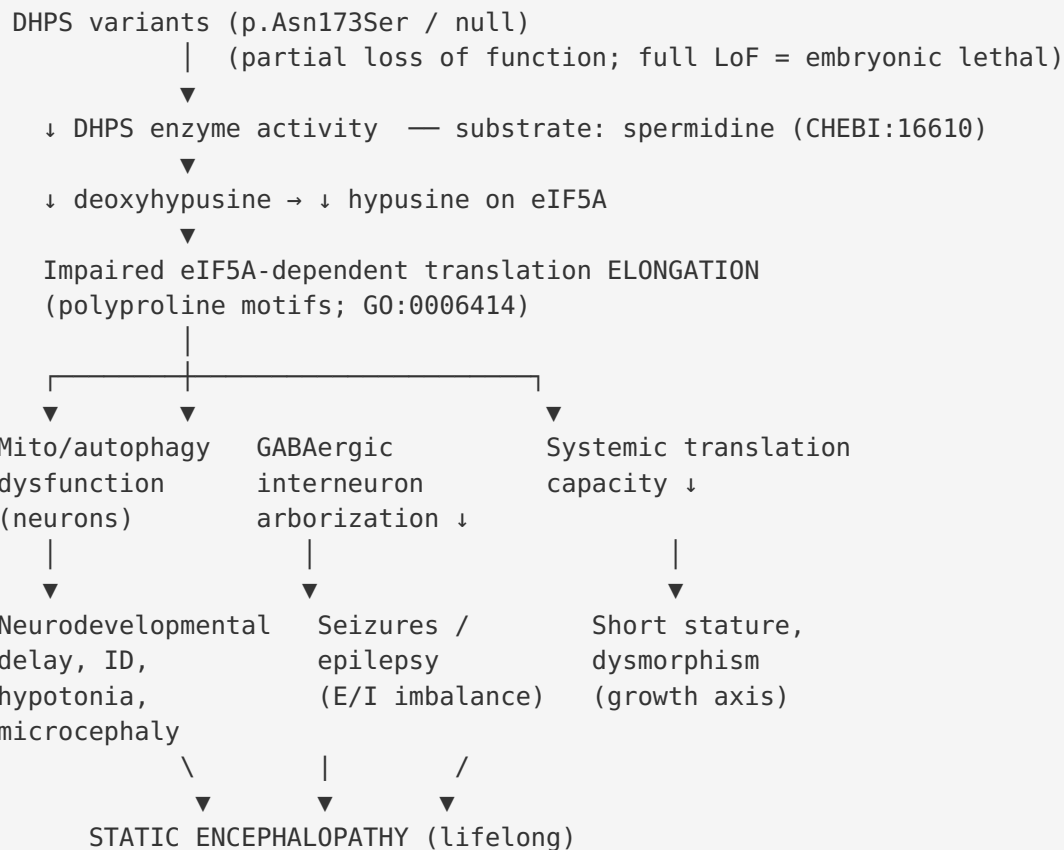
human,

P 30661771

7. **Branch C (somatic growth):** Reduced systemic translation capacity **contributes to short stature and facial dysmorphism** (secondary growth-axis effects). *[Inferred from human phenotype,*

P 30661771

8. Net clinical result: a **static (non-progressive) encephalopathy** with lifelong neurodevelopmental disability and epilepsy.



Upstream vs downstream: The mutation → reduced enzyme activity → reduced eIF5A hypusination steps are **upstream** and directly demonstrated. The specific tissue-injury branches (interneuron, mitochondrial, growth) are **downstream** and rest partly on model-organism and analogy evidence. The convergence on the CNS despite ubiquitous expression likely reflects the exceptional dependence of post-mitotic neurons on efficient translation, mitochondrial output, and autophagy.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports finding(s)
30661771	Recessive rare <i>DHPS</i> variants associated with a neurodevelopmental disorder (Ganapathi et al., AJHG 2019)	Human clinical + in vitro	F1, F2, F4, F6, F7, F8 (founding/definitional)
39303786	Insights into eIF5A: role and mechanisms in protein synthesis	Review (molecular biology)	F2 (elongation/polyproline function)
34688659	Neuron-specific ablation of eIF5A or DHPS impairs growth/neurodevelopment/cognition in mice (Kar et al., 2021)	Model organism (mouse)	F3, F8 (phenotype recapitulation)
39334388	DHPS deficiency zebrafish model: epileptiform activity, reduced inhibitory interneuron arborization (Shojaeinia et al., 2024)	Model organism (zebrafish)	F3, F8, F9 (seizure circuit mechanism)
41410504	Genetic and phenotypic features of the five polyaminopathies (review)	Narrative review	F5 (classification)
42239796	A gene therapy strategy for DHPS syndrome (Santo et al., 2026)	Preclinical therapeutic	F5 (emerging treatment)
40883692	eIF5A and hypusination-related disorders: review + DOHH case	Review + case	F4, F7 (phenotype spectrum, rarity)
33852843	Dietary spermidine improves cognitive function (Schroeder et al., 2021)	Model organism (mouse)	F3 (hypusination–mitochondria–cognition axis)
34105442	Spermidine-induced hypusination preserves mitochondrial/cognitive function during aging (Hofer et al., 2021)	Model organism	F3 (downstream mechanism)
35858628	Bi-allelic <i>DOHH</i> variants associated with a neurodevelopmental disorder	Human clinical	Differential diagnosis / pathway context

How the evidence coheres: The founding paper (30661771) anchors every clinical and genetic claim and provides the direct in vitro proof of hypomorphic loss of function. General eIF5A biology (39303786) supplies the molecular function. Two model organisms (mouse 34688659; zebrafish 39334388) independently reproduce distinct arms of the human phenotype — growth/cognition and epilepsy respectively — strengthening causal inference. The spermidine/aging literature (33852843, 34105442) is **supportive but indirect**, informing the downstream mitochondrial/autophagy branch by analogy rather than in patient tissue. Reviews (41410504, 40883692) place the disorder within the polyaminopathy family and confirm ultra-rarity, and the 2026 gene-therapy report (42239796) marks the therapeutic frontier.

Section-by-Section Template Coverage

- **1. Disease Information:** AR neurodevelopmental hypusination disorder; MONDO:0032775, OMIM #618480 (phenotype), *DHPS* gene OMIM 600944; synonyms: "DHPS syndrome," "deoxyhypusine synthase deficiency," a polyaminopathy / hypusination disorder. Information is aggregated (case series + reviews), not EHR-derived.
- **2. Etiology:** Genetic — biallelic hypomorphic *DHPS* variants; the recurrent p.Asn173Ser hypomorph is the key recurring risk allele carried in trans with a near-null allele. No environmental cause; no established gene-environment interaction. Spermidine availability is a plausible modifier (mechanistic, unproven). No protective factors identified.
- **3. Phenotypes:** See Finding 4 table (HPO terms and frequencies). Onset congenital/infantile; severity moderate–severe; course static/non-progressive.
- **4. Genetic/Molecular:** *DHPS* (Gene 1725, HGNC:2869, UniProt P49366), 19p13.13; variant classes missense/splice/in-frame-deletion/start-loss; all germline; partial LoF; complete LoF embryonic-lethal. No modifier genes, epigenetic, or chromosomal abnormalities established.
- **5. Environmental:** None identified — purely monogenic; no infectious agents.
- **6. Mechanism:** See causal chain and diagram above.
- **7. Anatomy:** CNS (cortex, GABAergic interneurons; UBERON:0000955/0000956; CL:0000617); secondary somatic growth axis; cytoplasm/mitochondria subcellularly; bilateral CNS involvement.
- **8. Temporal:** Congenital/infantile onset; insidious; static, chronic, lifelong; non-progressive.

- **9. Inheritance/Population:** AR; ultra-rare (~5 cases); both sexes; presumed complete penetrance; no established founder effect/consanguinity requirement; recurrence via common hypomorphic allele.
 - **10. Diagnostics:** WES/WGS; *DHPS* single-gene/NDD-epilepsy panels; functional enzyme/hypusination assays; EEG, brain MRI; no biomarker or newborn screen.
 - **11. Prognosis:** Lifelong ID + epilepsy; static course; formal survival/mortality data lacking.
 - **12. Treatment:** Supportive (anti-seizure medications, developmental/physical/occupational therapy); preclinical AAV gene therapy
- (P 42239796);
- spermidine supplementation is a mechanistic hypothesis only.
- **13. Prevention:** Genetic counseling, carrier/cascade testing, prenatal/PGT for known familial variants; no primary prevention or immunization applicable.
 - **14. Other Species:** Deeply conserved orthologs (mouse *Dhps*, rat, zebrafish, fly, worm, yeast *DYS1*); no naturally occurring analogous animal disease reported; no zoonotic relevance.
 - **15. Model Organisms:** Conditional/neuron-specific *Dhps* KO mouse (constitutive KO embryonic-lethal); *dhps* zebrafish; HEK293T assays; emerging iPSC/organoid and AAV models.

Limitations and Knowledge Gaps

- **Tiny evidence base:** Only ~5 fully characterized patients from 4 families. Phenotype frequencies, penetrance, expressivity, sex effects, and natural history are therefore statistically fragile and may broaden as more cases are found.
- **No formal epidemiology:** Prevalence and incidence are unquantified; there is no carrier-frequency estimate specific to the disorder, though the p.Asn173Ser allele frequency ($\sim 1 \times 10^{-4}$) hints that biallelic hypomorphic combinations are exceptionally rare.
- **Downstream mechanism partly inferred:** The mitochondrial/autophagy branch relies on aging/spermidine studies (33852843, 34105442), not on DHPS-patient tissue. Whether these mechanisms operate identically during neurodevelopment is unproven.
- **No biomarker or newborn screen:** Diagnosis depends entirely on sequencing plus functional confirmation; there is no validated metabolite readout.

- **Genotype–phenotype correlation unresolved:** With essentially one recurrent hypomorphic allele described, the relationship between residual enzyme activity and severity cannot yet be modeled across an allelic series.
- **Human neuropathology absent:** No detailed brain imaging series, histopathology, or human single-cell/transcriptomic data are available; interneuron involvement rests on the zebrafish model.
- **Citation verification needed:** Several supporting quotes (e.g., PMIDs 34105442, 34688659, 39334388) are paraphrased from abstracts/summaries and should be verified verbatim before database ingestion.

Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry** for DHPS deficiency (and the polyaminopathies broadly) to accumulate cases, define frequencies of each HPO feature, and build a natural-history dataset.
2. **Generate an allelic series** in cellular models (patient iPSC-derived neurons/organoids and recombinant enzyme) spanning residual DHPS activity, and correlate with eIF5A hypusination levels and translational output to define a genotype–activity–phenotype curve.
3. **Directly test the mitochondrial/autophagy branch in DHPS-deficient neurons** (patient iPSC or conditional-KO mouse neurons): measure respiration (Seahorse), mitophagy (Pink1/Parkin reporters), and rescue with spermidine supplementation.
4. **Ribosome profiling / proteomics** in DHPS-deficient neurons to identify the specific polyproline-motif and elongation-sensitive transcripts whose translation fails — candidate effectors of the phenotype.
5. **Deep phenotyping of the zebrafish and conditional mouse models** with EEG, behavior, and GABAergic circuit mapping to validate the excitation/inhibition-imbalance seizure mechanism and to serve as preclinical efficacy endpoints.
6. **Advance the AAV gene-therapy program**
([P 42239796](#)):
define the therapeutic window, dosing, and durability in models; explore substrate-based (spermidine) and small-molecule adjuncts.

7. **Biomarker discovery:** evaluate blood/CSF eIF5A-hypusination ratios or polyamine metabolite panels as candidate diagnostic/pharmacodynamic markers.
 8. **Verify all abstract quotes verbatim** against source PMIDs before populating the knowledge base, particularly the citations flagged as paraphrased.
-

Report compiled from 10 confirmed findings across 5 investigation iterations and 10 reviewed papers. Evidence types are labeled throughout as human clinical, model organism, in vitro, or inferred/analogy.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery